

BEHAVIORAL INEQUALITY: THE CONTRIBUTION OF DECISION-MAKING FRICTIONS TO INEQUALITY

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Preliminary and Incomplete 😊

WHAT WE DO

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 - Suboptimal refinancing, not claiming benefits, dominated insurance, ...

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- Approach #2: Combine Approach #1 with a **life cycle model**
 - ✓ Accounts for dynamic effects and endogenous responses
 - ⇒ Estimates of effects on aggregate consumption and wealth inequality

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Takeaway: Effects of frictions are **large**, **heterogeneous**, and **regressive**

WHAT DO WE MEAN BY A “FRICTION”?

Our Definition of a Friction

A **friction** is a choice made by an individual that differs from the choice that would be made by a hypothetical individual in identical circumstances that optimizes normative utility with full information and no non-monetary decision costs.

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- 2 We **do not** specify the underlying microfoundation(s)

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- Evaluate specific policies
- Measure the effects of specific biases
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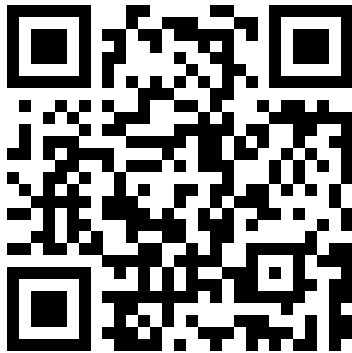
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- ③ If choice is not a friction, estimates are lower bound on utility costs to rationalize it

LIST OF FRICTIONS THAT WE CONSIDER

No.	Category	Friction	Syllabi Covering Friction		
			Top Universities (Out of 10)	High Schools (Out of 10)	Coverage in Top Journals
1	Education	Not Attending College Due to Frictions	0	10	.
2	Education	Attending 2-Year For-Profit Colleges	0	1	.
3	Savings	Not Meeting the Employer 401(k) Match	3	3	.
4	Savings	Not Investing in Stock Mutual Funds	10	10	.
5	Savings	Saving in Low Interest Accounts	5	4	.
6	Savings	Investing in Individual Stocks	6	7	.
7	Savings	Investing in High-Fee Mutual Funds	4	1	.
8	Borrowing	Not Refinancing a Mortgage	6	2	.
9	Borrowing	Not Getting an Additional Mortgage Quote	1	0	.
10	Borrowing	Suboptimally Repaying Credit Card Debt	0	0	.
11	Borrowing	Paying Late Fees	2	8	.
12	Borrowing	Paying Overdraft Fees	4	6	.
13	Insurance	Choosing Dominated Health Insurance	0	2	.
14	Insurance	Suboptimally Lapsing Life Insurance	1	0	.
15	Benefits	Not Claiming EITC Benefits	1	1	.
16	Benefits	Not Claiming SNAP Benefits	0	0	.
17	Health	Not Taking Statin Medications	0	0	.
18	Health	Not Screening for Colon Cancer	0	0	.
19	Health	Not Quitting Smoking	0	0	.



`timdesilva.me/frictions`

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- 1 Empirical Framework and Examples (Approach #1)
- 2 Results (Approach #1)
- 3 Life Cycle Model (Sketch) and Results (Approach #2)
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 - Groups: 25%–50%–25% splits of age-adjusted pre-tax pre-transfer income
- $s_{it} \equiv \mathbf{1}$ (i at risk of friction at t), $f_{it} \equiv \mathbf{1}$ (i subject to friction at t)
 - Example: $s_{it} = 1$ denotes EITC eligibility, $f_{it} = 1$ denotes non-claiming
 - f_{it} may depend on past choices (e.g. for-profit college attendance)

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- Budget constraint:

$$\text{Consumption}_{it}(f_{it}) + \underbrace{\overbrace{\text{Wealth}_{it+1}(f_{it})}^{\text{carried out of } t} - \overbrace{\text{Wealth}_{it}(f_{it})}^{\text{carried into } t}}_{\text{net savings}} = \underbrace{R_{it}(f_{it})}_{\text{"consumable" income}}$$

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- Consumption and wealth not observed, so construct consumable income directly:

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- Different frictions create losses through different channels

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Attending 2-Year For-Profit Colleges: Reduces Labor Income

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Not Investing in Stock Mutual Funds: Reduces **Capital Income**

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Not Claiming EITC Benefits: Reduces **Benefits**

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Paying Credit Card Late Fees: Creates **direct costs** from additional fees

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Not Quitting Smoking: Creates **direct costs** from cigarettes and increases **mortality**

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- Different frictions create losses through different channels
- **Challenge:** fully accounting for interaction between channels + dynamic effects
 - Ignore dynamics in this analysis, and incorporate in **life cycle model**

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Unconditional Loss: $\Delta_j \equiv \frac{\mathbb{E}[f_{it} l_{it}(f_{it}) \mid j_{it} = j]}{\mathbb{E}[A_{it}(f_{it}) \mid j_{it} = j]} = S_j F_j L_j$

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- Important parameters in life cycle model, but not needed for empirical approach:
 - **Correlation** of different frictions within-individual (in progress)
 - **Autocorrelation** of frictions within individual (for now: permanent)
 - **Sophistication** about frictions (for now: full sophistication)
- Caveats to both of our approaches:
 - Do not capture spillovers, GE effects, or **cross-subsidies** (could potentially incorporate)
 - Removing a particular (set of) friction(s) may not be **feasible** given microfoundations

EXAMPLES

EXAMPLE: PAYING LATE FEES

- s_{it} = having a credit card, f_{it} = paying at least one late fee
 - Shares S_j and F_j measured by income group (CFPB Making Ends Meet)
 - Adjust for survey underreporting of credit cards (Karlan-Zinman 2008)
- Financial impact comes from λ_{it} = direct losses from paying late fees
 - No micro data on fees $\Rightarrow$ estimate one L_j for all j (CFPB CCM Report)
 - $L_j = (\text{average late fee}) \times (\# \text{ fees per card} \mid \text{fees} > 1) \times (\# \text{ cards per cardholder})$

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Income Group (j)	S_j	F_j	L_j	Δ_j
Low Income	0.463 (0.013)	0.311 (0.017)	\$406 [381, 425]	\$57 [49, 64]
Middle Income	0.736 (0.004)	0.251 (0.010)	\$406 [381, 425]	\$73 [70, 78]
High Income	0.827 (0.007)	0.133 (0.011)	\$406 [381, 425]	\$44 [38, 50]

EXAMPLE: ATTENDING 2-YEAR FOR-PROFIT COLLEGES

- s_{it} = attending college, f_{it} = attending 2-year for-profit
 - Shares S_j and F_j measured by parental income group (BPS 12/17, Snyder et al. 2019)
 - Translate to child income group using intergenerational correlation (Haider-Solon 2006)
- Financial impact combines existing literature (Cellini-Turner 2019) + simulation
 - 1.5pp increase in unemployment probability $\Rightarrow \downarrow$ Labor Income
 - 11.3% earnings loss conditional on employment $\Rightarrow \downarrow$ Labor Income
 - \$6,100 increase in student debt (repaid under default plan) $\Rightarrow \lambda_{it}$

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Low Income	0.263 (0.016)	0.236 (0.014)	\$1,364 [1,116, 1,613]	\$85 [64, 106]
Middle Income	0.277 (0.010)	0.094 (0.007)	\$2,755 [1,961, 3,550]	\$72 [47, 97]
High Income	0.138 (0.012)	0.037 (0.011)	\$1,781 [447, 3,116]	\$9 [-14, 32]

EXAMPLE: NOT INVESTING IN STOCK MUTUAL FUNDS

- s_{it} = have financial wealth $\geq W_- \equiv \min\{2 \text{ mo. salary}, \$2K\}$ (SCF, Sabat-Gallagher 2019)
- f_{it} = not investing any wealth in diversified stocks (SCF)
 - Unlikely to be reflective of non-standard preferences (Choukhmane-de Silva 2026)
- Financial impact comes from $\downarrow$ Capital Income
 - $\ell_{it}(f_{it}) = \text{counterfactual equity share} \times (\text{financial wealth} - W_-)$
 - Counterfactual equity share = average age-specific equity share of those with $f_{it} = 0$

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Income Group (j)	S_j	F_j	L_j	Δ_j
Low Income	0.326 (0.008)	0.646 (0.015)	\$354 [297, 436]	\$75 [61, 92]
Middle Income	0.643 (0.006)	0.305 (0.008)	\$959 [887, 1,039]	\$188 [171, 206]
High Income	0.867 (0.006)	0.129 (0.007)	\$2,615 [2,193, 3,076]	\$293 [239, 354]

EXAMPLE: NOT QUITTING SMOKING

- s_{it} = smoke before 50 and want to quit, f_{it} = want to quit before 50 but don't (NHIS)
- Choose age 50 as critical year for two reasons:
 - 1 Those who successfully quit before 50 have similar mortality to never-smokers
 - 2 Those who quit after 50 have similar mortality to non-smokers
- Financial impact comes from **mortality** and **direct losses**
 - Simulate increased ($\sim 2x$) mortality prob. by income-age $\Rightarrow A_{it}(0) - A_{it}(f_{it}) \geq 0$
 - Cost of cigarettes $\sim \$1.2K/\text{year} \Rightarrow \lambda_{it}$

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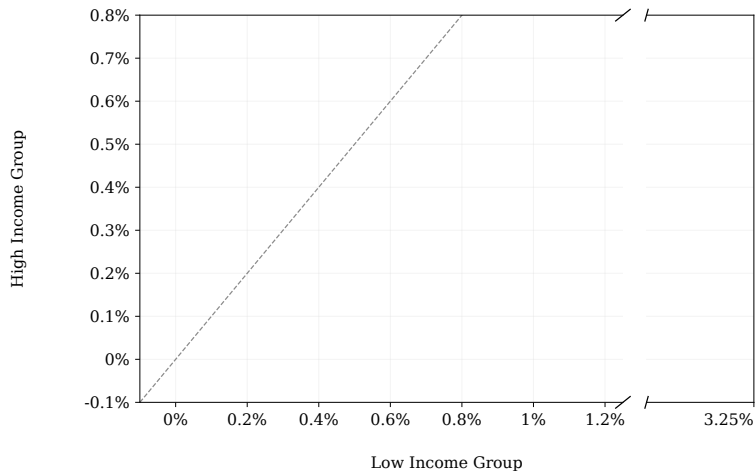
Income Group (j)	S_j	F_j	L_j	Δ_j
Low Income	0.414 (0.027)	0.563 (0.031)	\$2,853 [2,636, 3,071]	\$665 [530, 801]
Middle Income	0.350 (0.020)	0.414 (0.028)	\$3,599 [3,190, 4,008]	\$521 [396, 647]
High Income	0.298 (0.024)	0.240 (0.047)	\$5,211 [2,342, 8,081]	\$373 [144, 602]

- 1 Empirical Framework and Examples (Approach #1)
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- 4 Conclusion

FINANCIAL IMPACTS AS % OF PERMANENT CONSUMABLE INCOME

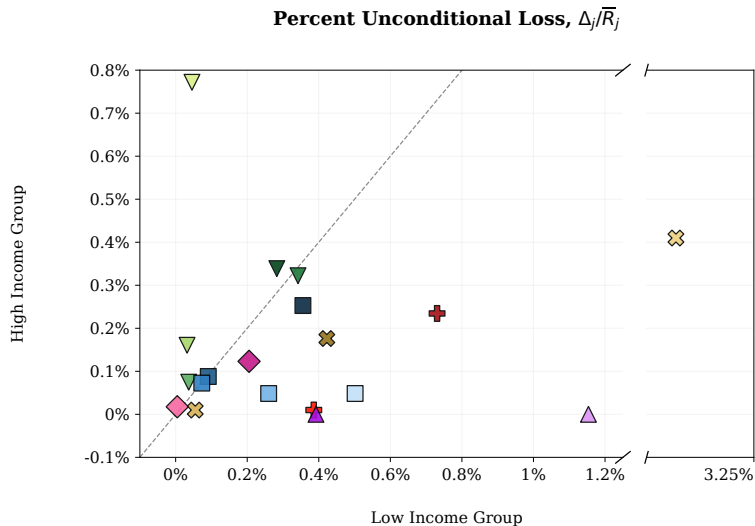
Percent Unconditional Loss, $\Delta_j/\bar{R}_j$

---- y = x (Equality)



► Dollars

FINANCIAL IMPACTS AS % OF PERMANENT CONSUMABLE INCOME



---- y = x (Equality)

Education

- ✚ Not Attending College Due to Frictions
- ✚ Attending 2-Year For-Profit Colleges

Savings

- ▼ Not Meeting the Employer 401(k) Match
- ▼ Not Investing in Stock Mutual Funds
- ▼ Saving in Low Interest Accounts
- ▼ Investing in Individual Stocks
- ▼ Investing in High-Fee Mutual Funds

Borrowing

- Not Refinancing a Mortgage
- Not Getting an Additional Mortgage Quote
- Suboptimally Repaying Credit Card Debt
- Paying Late Fees
- Paying Overdraft Fees

Insurance

- ◆ Choosing Dominated Health Insurance
- ◆ Suboptimally Lapsing Life Insurance

Benefits

- ▲ Not Claiming EITC Benefits
- ▲ Not Claiming SNAP Benefits

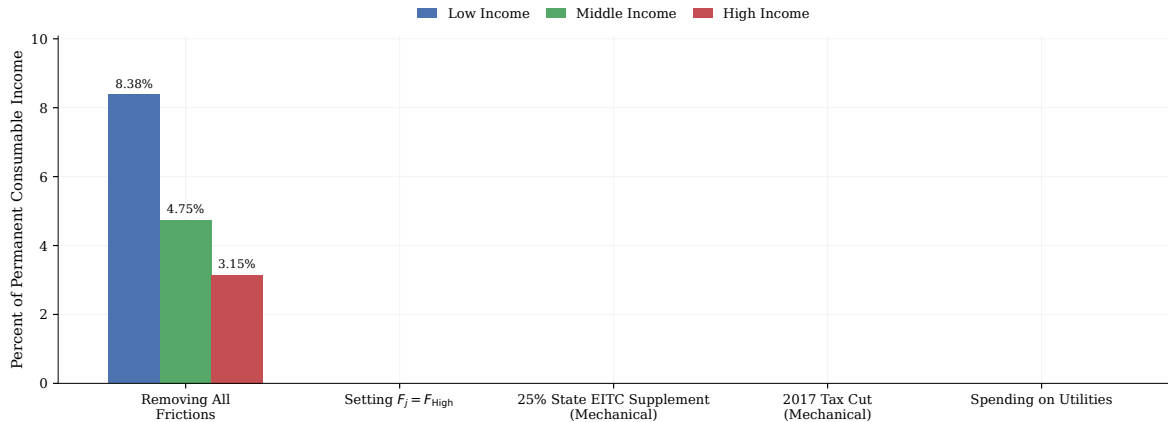
Health

- ✚ Not Taking Statin Medications
- ✚ Not Screening for Colon Cancer
- ✚ Not Quitting Smoking

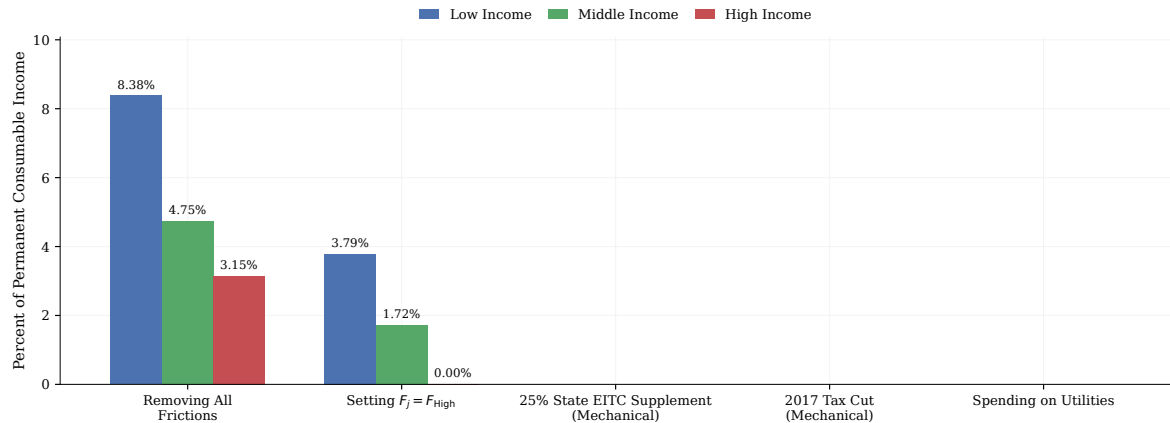
Income Group	Average	All
Low	0.44%	8.38%
High	0.17%	3.15%

► Dollars

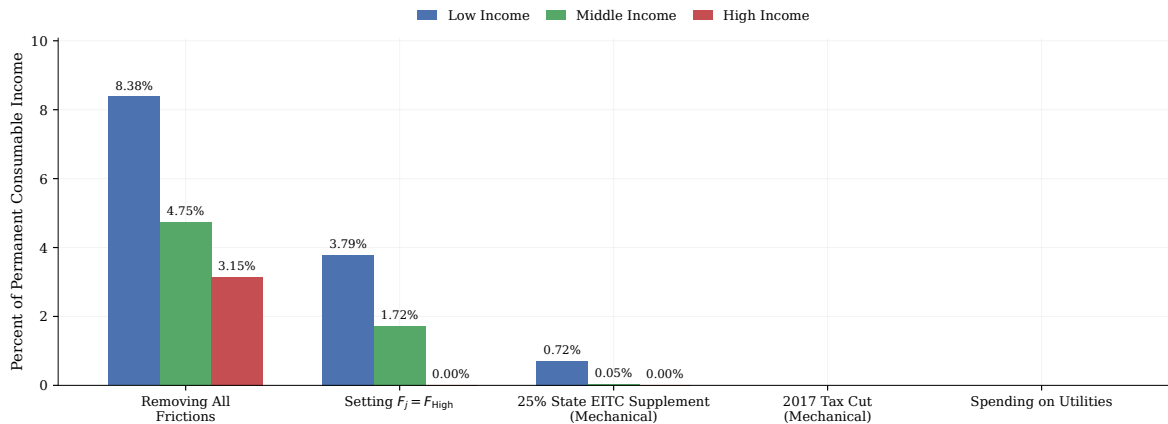
BENCHMARKING TOTAL LOSSES FROM FRICTIONS



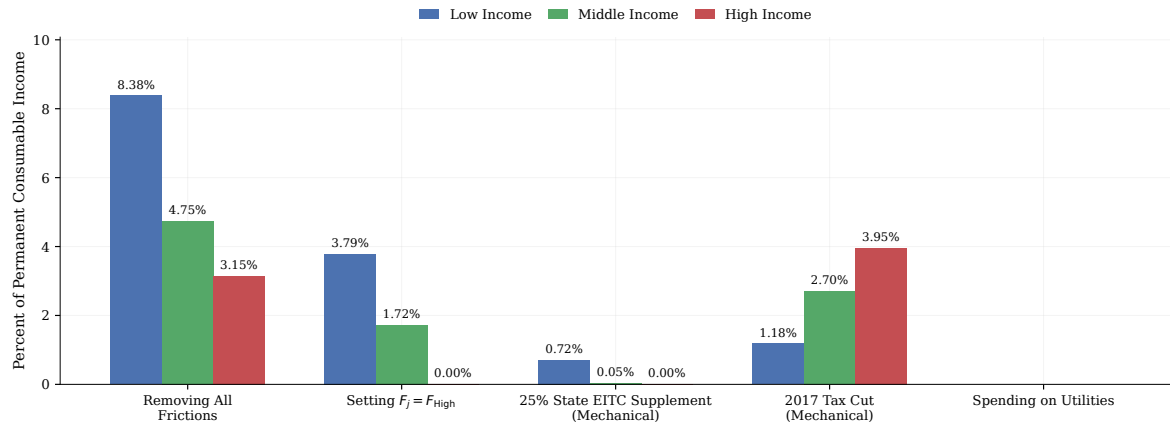
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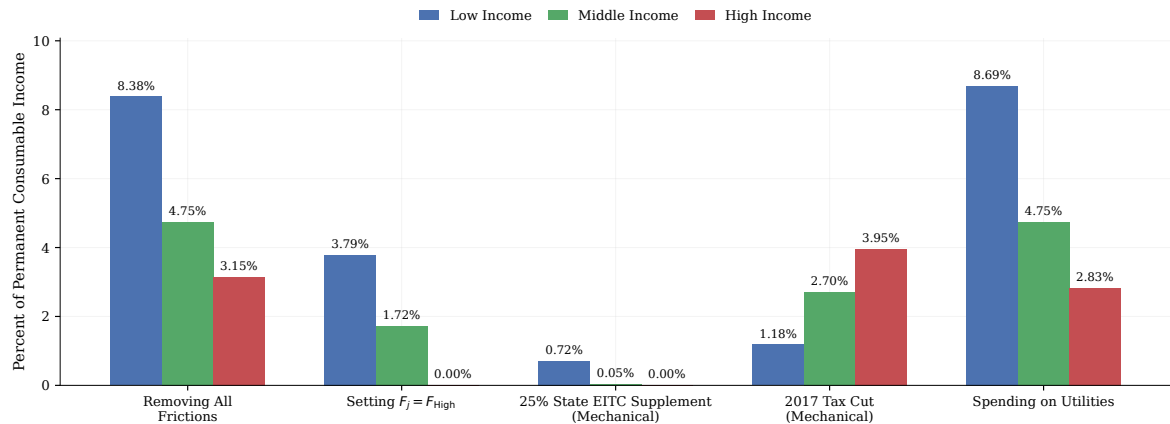
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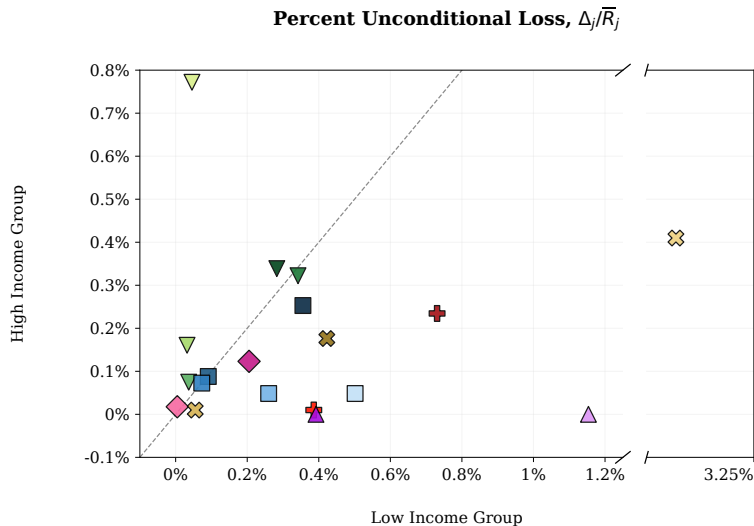
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BENCHMARKING TOTAL LOSSES FROM FRICTIONS



FINANCIAL IMPACTS: CATEGORY AVERAGES



---- y = x (Equality)

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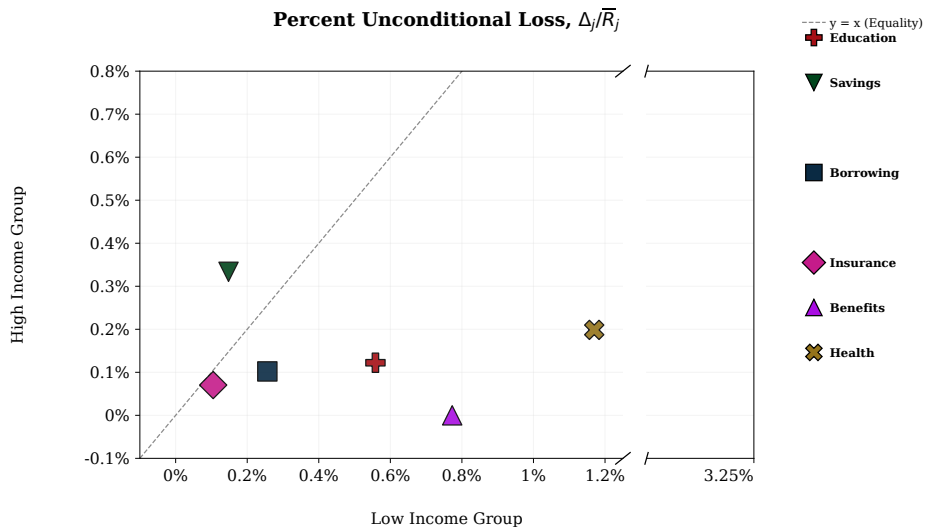
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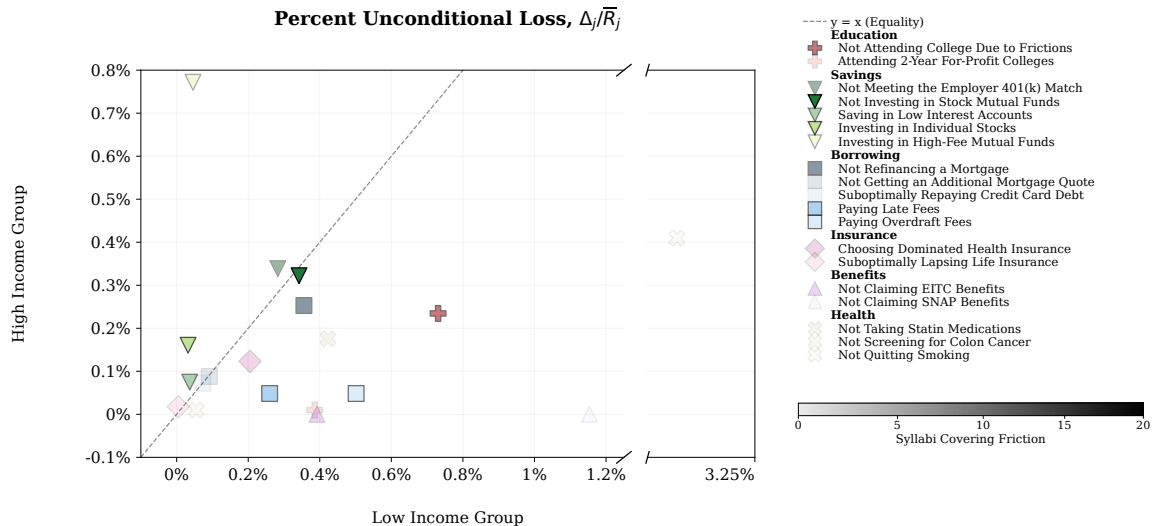
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FINANCIAL IMPACTS: CATEGORY AVERAGES

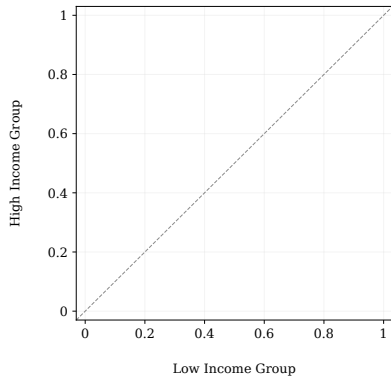


FINANCIAL IMPACT OF FRICTIONS: SYLLABUS COVERAGE

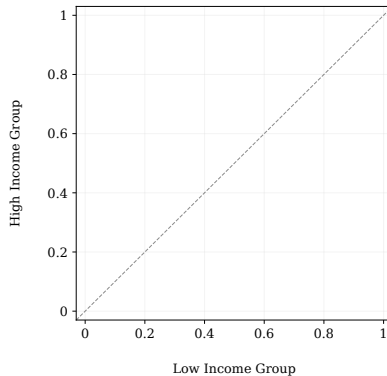


DECOMPOSING FINANCIAL IMPACT: SHARES AND PREVALENCE

Share at Risk, S_j



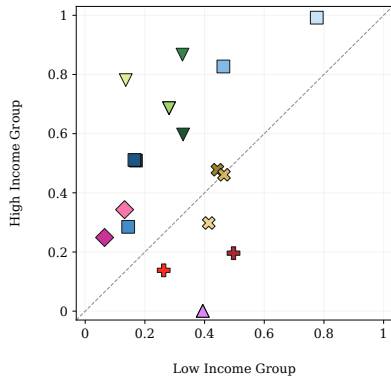
Friction Prevalence, F_j



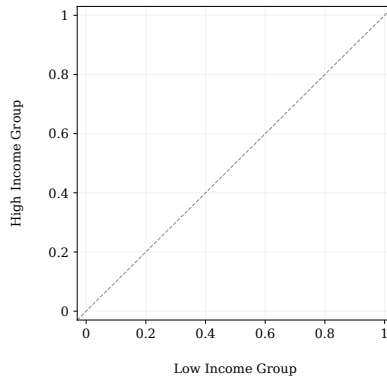
---- $y = x$ (Equality)

DECOMPOSING FINANCIAL IMPACT: SHARES AND PREVALENCE

Share at Risk, S_j



Friction Prevalence, F_j



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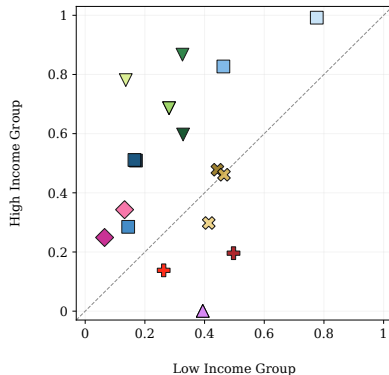
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Health

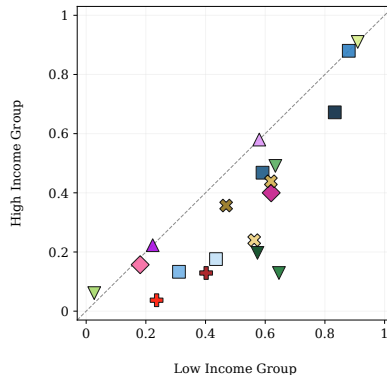
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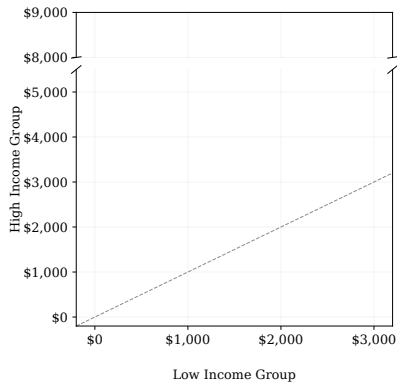
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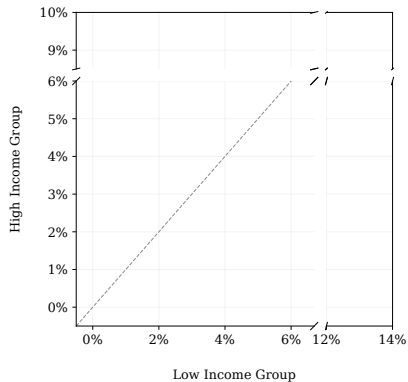
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DECOMPOSING FINANCIAL IMPACT: AVERAGE LOSSES

Average Loss, L_j

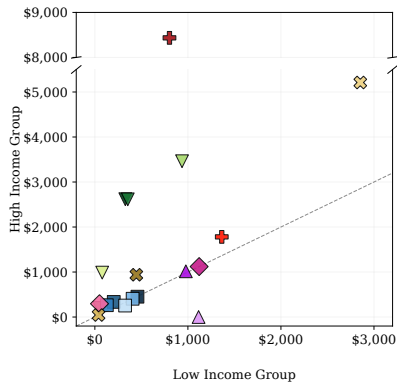


Percent Average Loss, $L_j/\bar{R}_j$

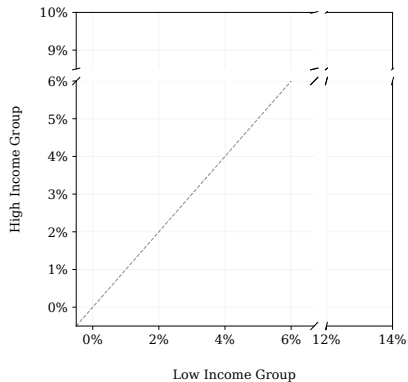


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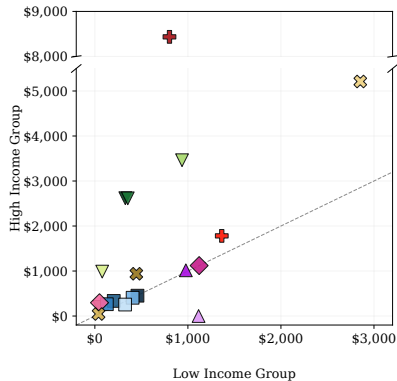
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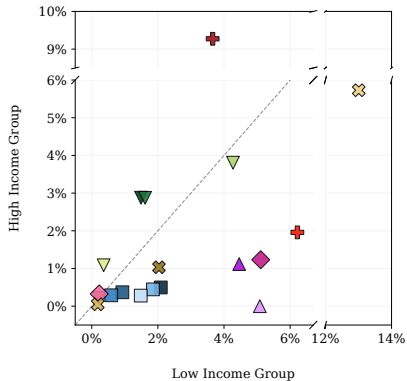
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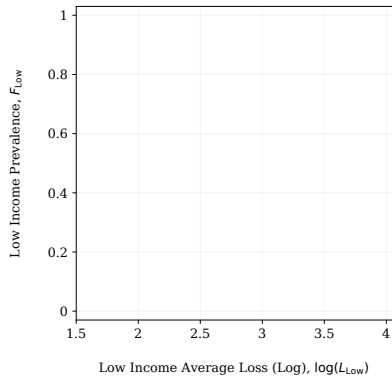
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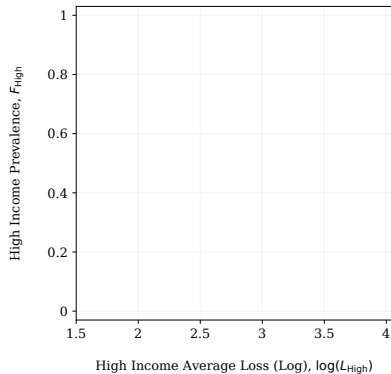
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RELATIONSHIP BETWEEN PREVALENCE AND AVERAGE LOSS

Low Income Group

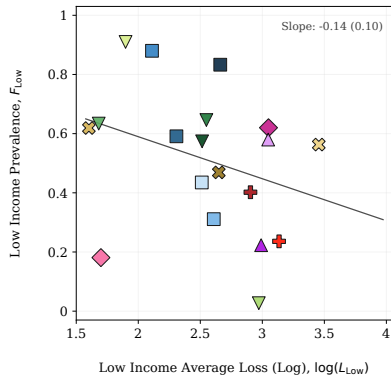


High Income Group

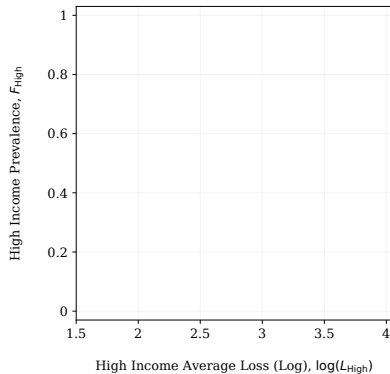


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Low Income Group



High Income Group



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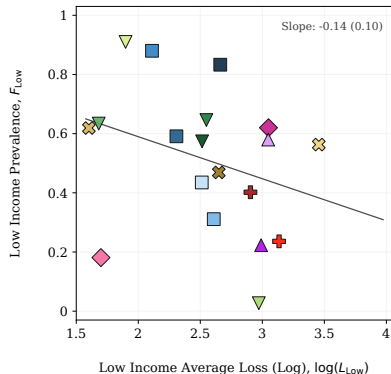
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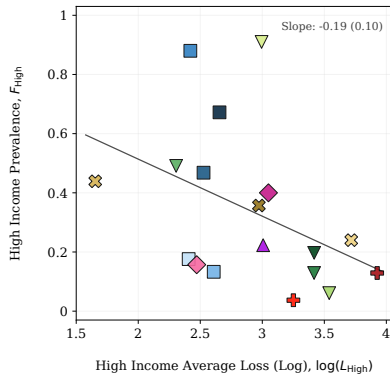
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- ① **Evolution** of key parameters over time
 - Losses seem to respond to policies (e.g. for-profit certifications, public health efforts)
- ② **Correlation** of prevalence across-frictions within-individual
 - Currently have estimated 6 bivariate correlations with mean 0.032 (SE 0.006)
- ③ Comparison of losses with **forecasts** from a survey of academics
- ④ Comparison of losses with coverage in **academic journals** and **media**

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... More ideas welcome!

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- 2 Results (Approach #1)
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- Reasons to develop a life cycle model:
 - ① Some frictions have **dynamic** effects (e.g. 401(k) match, stock market)
 - ② Removing a friction likely creates **endogenous** responses
 - ③ Study affects on aggregate **consumption** and **wealth** inequality

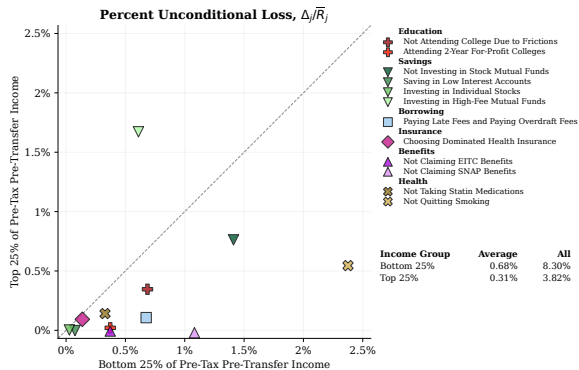
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 - Individuals ex-ante heterogeneous in permanent income and education
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- **Key counterfactuals:** Set $\mathbf{f}_i(n) = 0$ for each friction n and compare with data

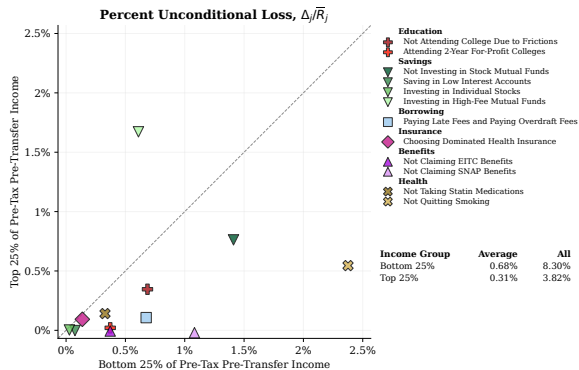
FINANCIAL IMPACT OF FRICTIONS: TWO APPROACHES

Life Cycle Model

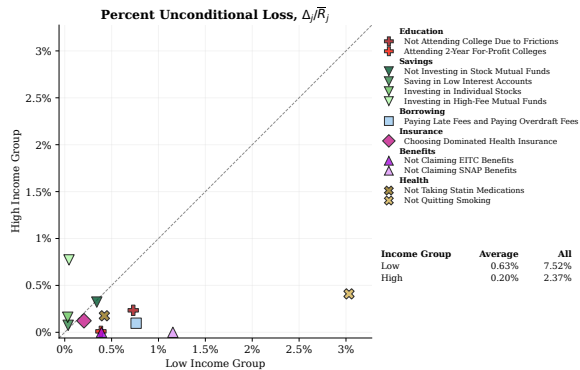


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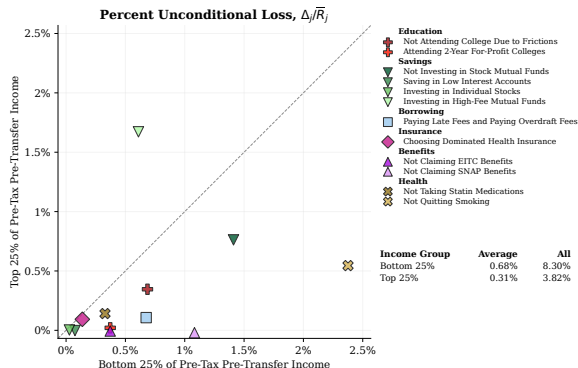


Empirical Estimates

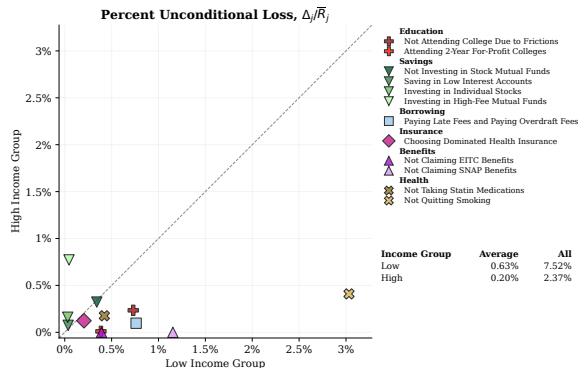


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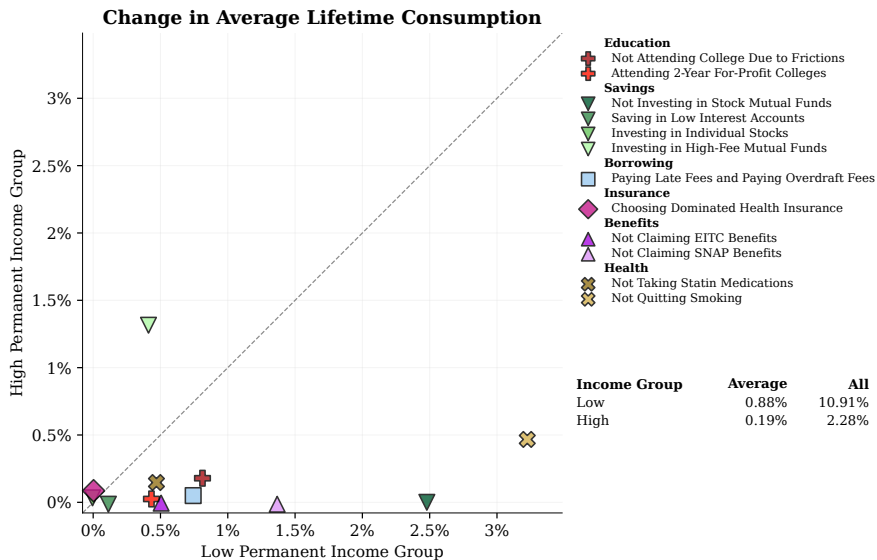


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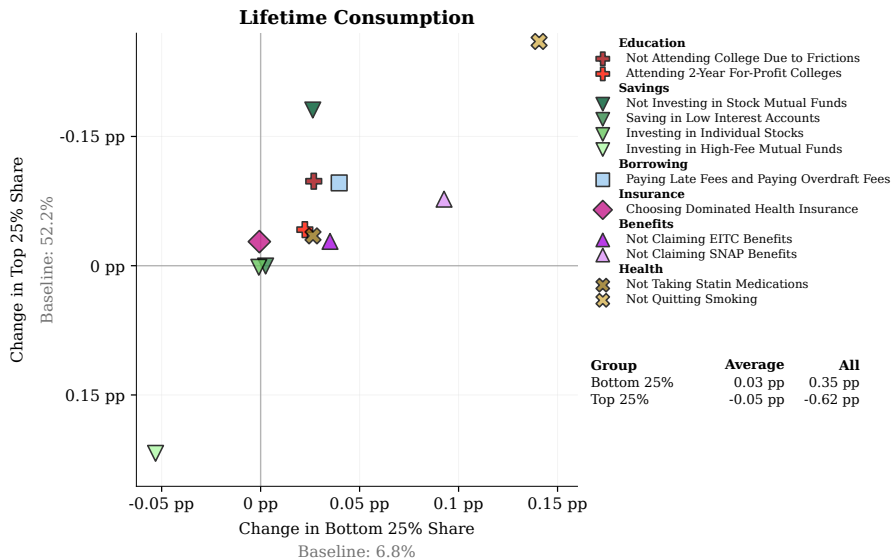


Losses are much larger for **savings** frictions, due to **dynamic** effects

FRICTIONS HAVE LARGE EFFECTS ON LIFETIME CONSUMPTION



FRICTIONS REDUCE LIFETIME CONSUMPTION INEQUALITY



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- ① Decision-making frictions create meaningful losses in aggregate
 - Aggregate losses are **8%** (**3%**) for low (high) income = expenditure on utilities
 - Largest (as of now): smoking, fund fees, college attendance, SNAP benefits
- ② Losses from decision-making frictions are **regressive**
 - High-income have more to lose (L), but low-income **prevalence** (F) is higher
 - Main exception: Savings frictions, which are most covered by syllabi
 - **Model:** Removing frictions decreases inequality in lifetime consumption

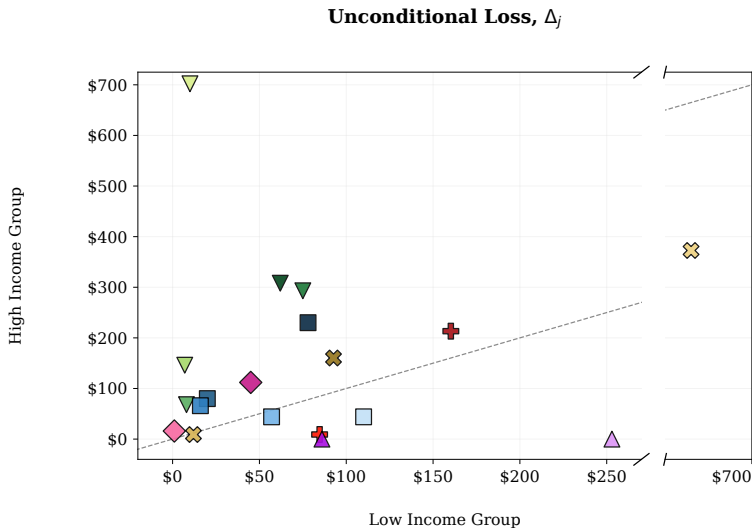
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... Much more to come!

THANK YOU!

`sdellavi@berkeley.edu`
`tdesilva@stanford.edu`
`rohanjha@g.harvard.edu`

FINANCIAL IMPACTS IN DOLLARS



---- y = x (Equality)

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Income Group	Average	All
Low	\$97	\$1,839
High	\$151	\$2,862

◀ Back

FINANCIAL IMPACTS: CATEGORY AVERAGES IN DOLLARS

